Do mobile health apps actually help people improve diet or lifestyle?

## Yes, **mHealth apps improve diet and lifestyle** — but effects are modest, evidence quality is low, and long-term sustainability remains unclear.

**Do mobile health apps help people improve diet or lifestyle?**

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

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The literature shows broad agreement that mobile health apps can produce measurable improvements in dietary quality, physical activity, weight management, and chronic disease indicators, though effect sizes are generally small and study quality is often limited.

FIGURE 1 Research consensus on mHealth app effectiveness for diet and lifestyle

### Evidence for Dietary Improvement

Mobile apps produce measurable gains in diet quality across diverse populations. A systematic review of ten studies found that six reported improved diet quality following app use, with improvements in overall diet quality scores or individual dietary assessments (Scarry et al., 2022). In a randomized controlled trial of adults with prediabetes, an mHealth-enabled lifestyle intervention improved Alternate Healthy Eating Index scores by 6.2 points and reduced sugar-sweetened beverage intake by 0.5 servings per day compared with standard care (Han et al., 2023). A systematic review of 34 RCTs found that internet-connected nutrition apps consistently encouraged healthier eating habits, including increased fruit and vegetable consumption and reduced salt, fat, and sugar intake (Seid et al., 2024).

A 2025 meta-analysis of 21 studies (12,898 participants) found that app use increased fruit and vegetable consumption by 0.48 portions per day and decreased meat consumption by 0.10 portions per day (Curtin et al., 2025). Apps with **messaging and personalization features** were more effective, producing 0.75 and 0.67 additional daily fruit and vegetable portions, respectively (Curtin et al., 2024). Among cardiac patients, an mHealth app significantly improved Mediterranean diet adherence and increased consumption of healthy foods while decreasing unhealthy foods compared with standard care (Bernal-Jiménez et al., 2024).

### Weight, Chronic Disease, and Physical Activity Outcomes

| Outcome Domain | Evidence Summary | Key Finding |
|---|---|---|
| Weight management | 58% of weight-focused app studies showed effective behavior change (Salas-Groves et al., 2022) | Apps reduced weight and improved glucose and blood pressure when incorporating self-monitoring and tailored feedback (Dounavi & Tsoumani, 2019) |
| Glycemic control | 69% of studies measuring biometric outcomes showed significant reductions (Salas-Groves et al., 2022) | Diabetes apps reduced HbA1c by ~0.5% on average, especially within integrated clinical care (Barwacz et al., 2025) |

| Outcome Domain | Evidence Summary | Key Finding |
| --- | --- | --- |
| Physical activity | 14 of 21 studies targeting activity showed significant improvements (Schoeppe et al., 2016) | Cardiac patients using an mHealth app reported 619 vs. 472 minutes of weekly activity versus control (Bernal-Jiménez et al., 2024) |
| Quality of life | mHealth nutrition apps significantly improved QOL in cancer populations (Salas-Groves et al., 2022) | Physical quality-of-life scores improved in cardiac patients using an app versus standard care (Bernal-Jiménez et al., 2024) |

FIGURE 2 mHealth app effectiveness across chronic disease and lifestyle outcomes

Multi-component interventions that pair apps with additional support appear more effective than stand-alone apps. A systematic review of 27 studies found that 62% of multi-component interventions showed significant between-group improvements compared with 36% of stand-alone app interventions (Schoeppe et al., 2016). Among 46 studies of nutrition apps in chronic disease populations, mHealth interventions significantly improved health outcomes, with effect sizes comparable to traditional nondigital interventions (Salas-Groves et al., 2022).

### Limitations and Caveats

- **Few studies found significant differences** between app and control groups in a 52-RCT review, leading the authors to conclude there was no strong evidence for app effectiveness on health behaviors or outcomes (Milne-Ives et al., 2019).
- A 2026 systematic review of healthy adults identified only two eligible RCTs with **low quality of evidence** due to high risk of bias and missing data, concluding that app effectiveness cannot be assessed (Leuzzi et al., 2026).
- Of 46 chronic disease studies, only 24% measured maintenance of behavior change; 36% of those showed **decline or discontinued use** over 6–12 months (Salas-Groves et al., 2022).
- Less than half of nutrition app studies were grounded in behavioral theory, and no single app functionality was statistically more effective than others (Salas-Groves et al., 2022; Pala et al., 2024).
- Overdependence on apps may promote **unhealthy eating behaviors** such as orthorexia, and users relying solely on apps may receive advice of variable scientific rigor (Scarry et al., 2022).
- Effectiveness is described as **moderate and variable** across meta-analyses, with challenges including data security, digital inequalities, and lack of standardization in app quality (Barwacz et al., 2025).

### Synthesis

The evidence supports a qualified yes: mobile health apps can improve diet quality, physical activity, weight management, and chronic disease indicators, particularly when they incorporate self-monitoring, tailored feedback, messaging, and personalization. However, effect sizes are modest, study quality is frequently low, long-term engagement declines in a substantial proportion of users, and the strongest effects appear in chronic disease populations rather than healthy adults. The literature consistently calls for higher-quality RCTs, longer follow-up periods, and better characterization of which app features and behavior change techniques drive sustained results.

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